Label Organization Categories

CSC 482 Natural Language Processing (Project 8)

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Project Overview

Given 3 datasets of about 60,000 organizations with their corresponding categories, our program accepts the user-inputted name of an organization and uses NLP techniques to guess at the categories across the industry, group, and business levels corresponding to that organization.

Our tech stack includes Python and other libraries like pandas and NLTK for the main program. Our app is hosted locally through React.

Demo

Our Process

Data Cleaning

Clean up the tables and combine them into one dataframe

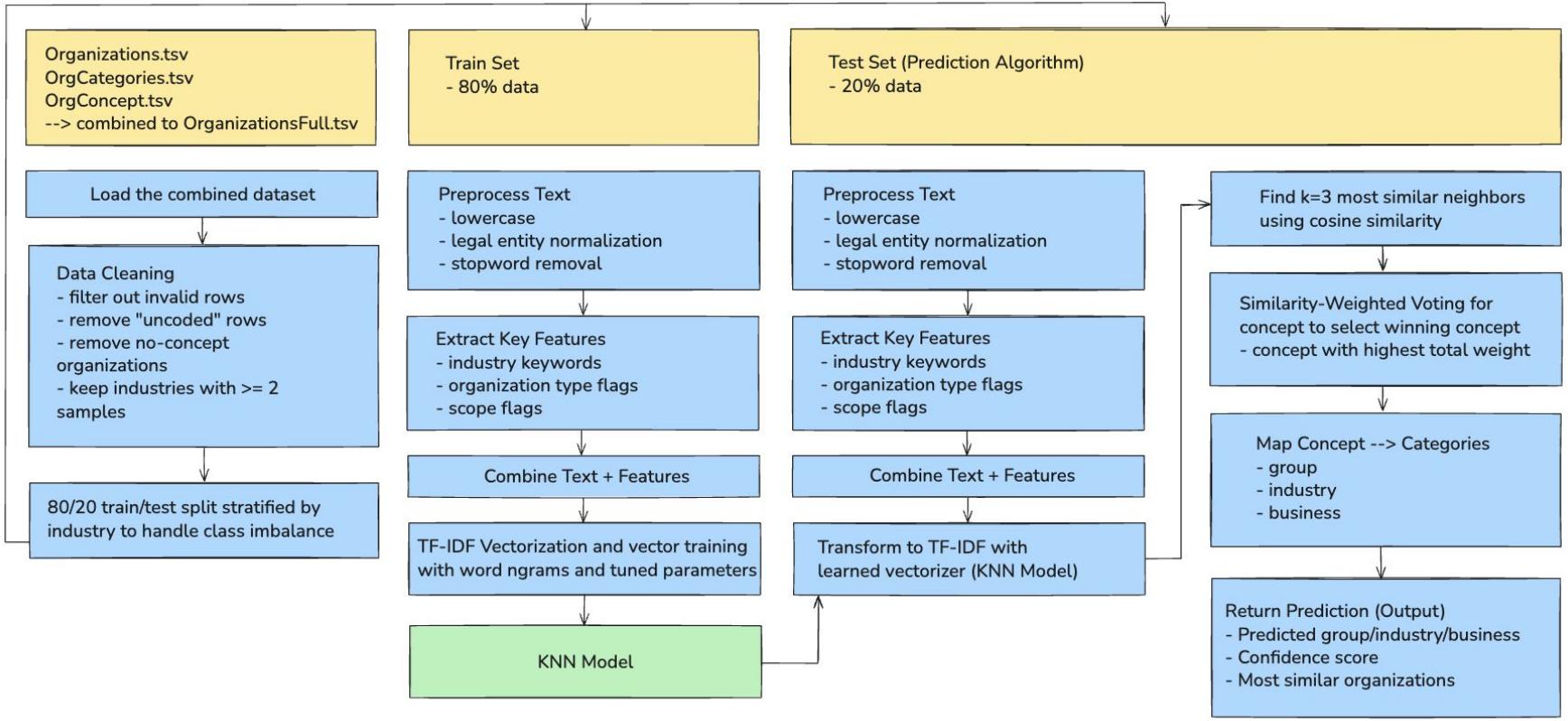
Data Preprocessing

Tokenization, normalization, feature extraction, TF-IDF vectorization

Base Model

Cosine similarity, kNN, weighted voting, concept mapping

Base Model


Organizations.tsv
OrgCategories.tsv
OrgConcept.tsv
--> combined to OrganizationsFull.tsv
Train Set
- 80% data
Test Set (Prediction Algorithm)
- 20% data
Load the combined dataset
Data Cleaning
- filter out invalid rows
- remove "uncoded" rows
- remove no-concept organizations
- keep industries with >= 2 samples
80/20 train/test split stratified by industry to handle class imbalance
Preprocess Text
- lowercase
- legal entity normalization
- stopword removal
Extract Key Features
- industry keywords
- organization type flags
- scope flags
Combine Text + Features
TF-IDF Vectorization and vector training with word ngrams and tuned parameters
KNN Model
Preprocess Text
- lowercase
- legal entity normalization
- stopword removal
Extract Key Features
- industry keywords
- organization type flags
- scope flags
Combine Text + Features
Transform to TF-IDF with learned vectorizer (KNN Model)
Find k=3 most similar neighbors using cosine similarity
Similarity-Weighted Voting for concept to select winning concept
- concept with highest total weight
Map Concept --> Categories
- group
- industry
- business
Return Prediction (Output)
- Predicted group/industry/business
- Confidence score
- Most similar organizations

Results: Base Model

	Business	Group	Industry
Accuracy	0.8090	0.8806	0.8320
Macro F1	0.6223	0.8568	0.6957

Overall (Avg) Accuracy: 0.8405
Mean Confidence: 0.7078

Our Process (cont.)

kNN Optimization

Test and determine the optimal k value

TF-IDF Optimization

Test and determine the optimal TF-IDF configuration

Web Interface

React web interface for usability

Results: Optimized Model

	Business	Group	Industry
Accuracy	0.8990	0.9522	0.9229
Macro F1	0.7800	0.9321	0.8419

Overall (Avg) Accuracy: 0.9247
Mean Confidence: 0.8721

Issues and Limitations

01 Data Leakage

Can’t classify organizations without a known concept ID

02 Training Knowledge

Can’t predict for organizations that don’t have a concept

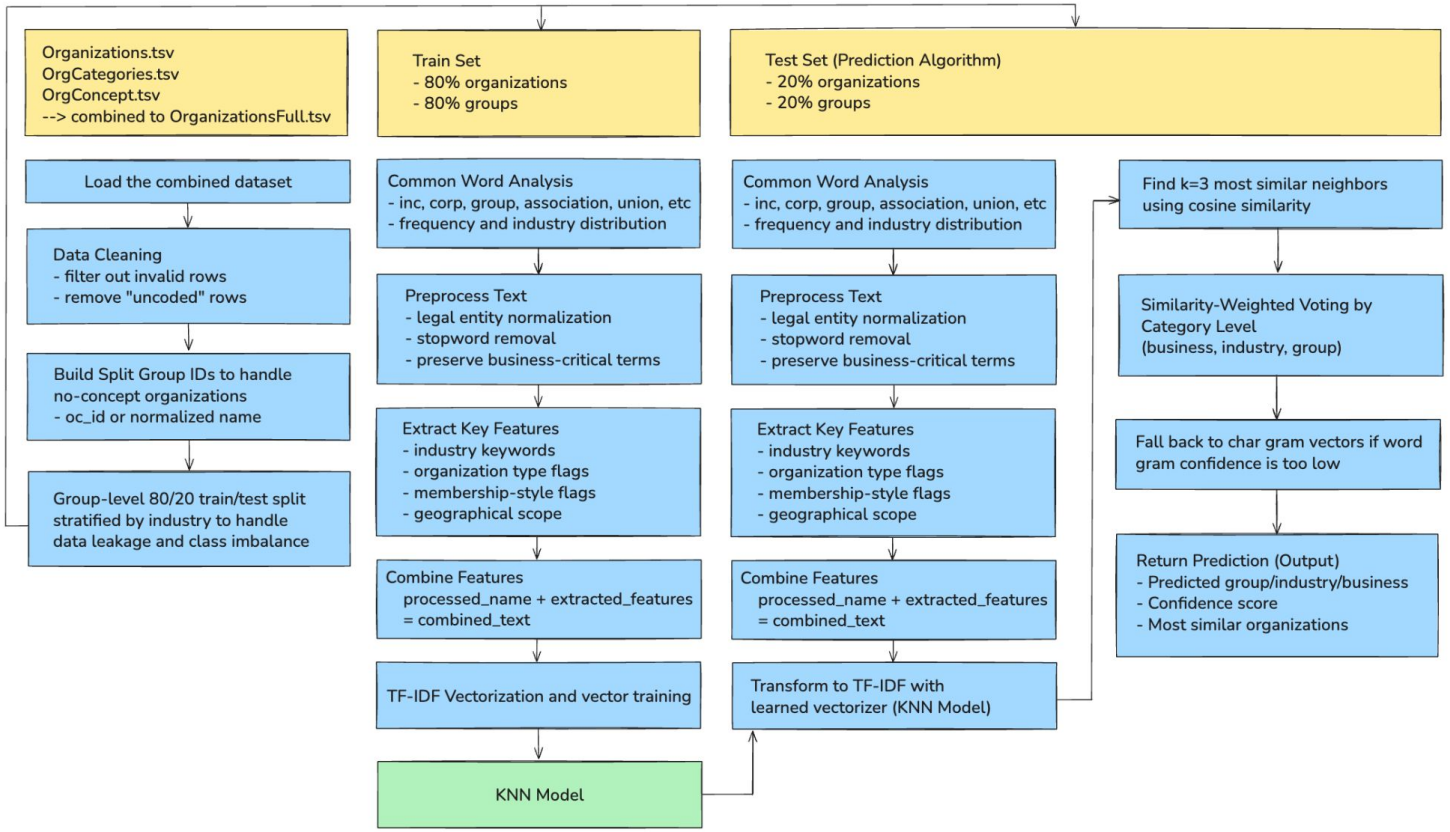
03 Feature Engineering

Not much distinction or analysis between common or abbreviated words

04 Concept Reliance

Rely on concepts to map to outputs instead of independently predicting each category

Revised Model



Results: Revised Model

	Business	Group	Industry
Accuracy	0.6420	0.8185	0.7352
Macro F1	0.5471	0.8014	0.6691

Overall (Avg) Accuracy: 0.7319
Mean Confidence: 0.8346

Key Technical Achievements

01 Domain-Specific Text Processing

Expanded common abbreviations, preserved important stopwords, etc.

02 Hybrid Similarity Model

Fallback if the initial model has low confidence

03 Explainable Weighted KNN

Utilized a weighted vote to improve KNN performance, added nearest neighbors to enhance explainability

04 Leakage Resistant Data Splitting

Split at organization-level instead of data-level

Thanks for listening!

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